VINOGRADOV'S CONJECTURES ON PRIMITIVE ROOTS AND CHARACTER SUMS

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ABSTRACT. For every $\eta > 0$ every prime p has a prime primitive root $\ell \ll_\eta (\tau(p-1)^2 \log p)^{8+\eta}$, so the least primitive root is $O_\varepsilon(p^\varepsilon)$. For every nonprincipal character χ modulo q, $\sum_{n \le x} \chi(n) \ll_\varepsilon q^\varepsilon x^{7/8+\varepsilon}$, which is $o(x)$ once $x \ge q^\eta$. Both were conjectured by Vinogradov. The proofs use the zero-free half-plane $\operatorname{Re} s > 7/8$ and are checked in Lean 4.

1. INTRODUCTION

For a prime p let $g(p)$ be the least primitive root modulo p and $g^*(p)$ the least prime primitive root. Vinogradov conjectured that $g(p) \ll_\varepsilon p^\varepsilon$ for every $\varepsilon > 0$. Burgess [3] proved $g(p) \ll_\varepsilon p^{1/4+\varepsilon}$; for the prime version the least prime in a progression [10] gives $g^*(p) \ll p^{5.5}$. Under the generalised Riemann hypothesis Shoup [17] and Martin [12] proved $g^*(p) \ll \omega^4(\log\omega + 1)^4 \log^2 p$ with $\omega = \omega(p-1)$. Erdős [7] asked whether every prime p has a prime $q < p$ that is a primitive root modulo p (Problem 985 of [2]).

Let τ denote the divisor function. Throughout, χ is a Dirichlet character and $L(s,\chi)$ its L-function.

Theorem 1.1. *For every $\eta > 0$ there is C_η such that every prime p has a prime primitive root ℓ with*

$$\ell \le C_\eta\big(\tau(p-1)^2 \log(p+2)\big)^{8+\eta}.$$

Corollary 1.2. *For every $\varepsilon > 0$ there is C_ε such that every prime p has a prime primitive root $\ell \le C_\varepsilon p^\varepsilon$. In particular $g(p) \le g^*(p) \ll_\varepsilon p^\varepsilon$.*

Corollary 1.3. *There is p_0 such that every prime $p \ge p_0$ has a prime $q < p$ that is a primitive root modulo p.*

Vinogradov also conjectured that $\sum_{n \le x} \chi(n) = o(x)$ for nonprincipal characters χ modulo q once $x \ge q^\varepsilon$; see [8]. For prime moduli Burgess [3] proved this for $x \ge q^{1/4+\varepsilon}$, and it follows from the generalised Riemann hypothesis.

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Theorem 1.4. *For every $\delta > 0$ there is C_δ such that for every nonprincipal character χ modulo q and every $s = \sigma + it$ with $\sigma \geq 7/8 + \delta$,*

$$|L(s,\chi)| + |L(s,\chi)|^{-1} \leq C_\delta\big(q(3+|t|)\big)^{\delta}.$$

Theorem 1.5. *For every $\delta > 0$ there is C_δ such that for every nonprincipal character χ modulo q and every $x \geq 1$,*

$$\Big|\sum_{n\leq x} \chi(n)\Big| \leq C_\delta\, q^{\delta} x^{7/8+\delta}.$$

Corollary 1.6. *For every $\eta > 0$ and $\theta > 7/8$ there is C such that $\big|\sum_{n\leq x}\chi(n)\big| \leq Cx^{\theta}$ for every nonprincipal character χ modulo q and every $x \geq q^{\eta}$. In particular $\sum_{n\leq x}\chi(n) = o(x)$ uniformly for $x \geq q^{\eta}$.*

The input is the zero-free half-plane of [14], from which [14] deduces, as in [1, Thm. 6.7], that the least quadratic nonresidue is $\ll_\varepsilon p^{\varepsilon}$. A primitive root must avoid every subgroup of prime index at once, and the number of characters involved is the source of the factor $\tau(p-1)^2$.

Theorem 1.7 ([14])**.** *If χ is a nonprincipal Dirichlet character, then $L(s,\chi) \neq 0$ for $\operatorname{Re} s > 7/8$.*

Theorem 1.7 is formalised in Lean 4 in [14]. Theorems 1.1, 1.4 and 1.5 and Corollaries 1.2, 1.3 and 1.6 are formalised here, with Theorem 1.7 as the only external input (Section 6). The constants are effective.

Theorem 1.4 is Littlewood's deduction of a Lindelöf bound from a zero-free half-plane [19, §14.2], and Theorem 1.5 follows from it by a contour shift; the new ingredient is Theorem 1.7. Among the consequences of Theorem 1.7, [14] lists the least quadratic nonresidue, the prime number theorem in progressions, class numbers and idoneal numbers, but not Theorems 1.4 and 1.5. All results here hold with $7/8$ replaced by any $\theta \in [1/2, 1)$ such that $L(s,\chi) \neq 0$ for $\operatorname{Re} s > \theta$ and every nonprincipal χ, for instance by the value $0.874957\ldots$ claimed in [11]. The companion papers [15, 16] prove, from a zero-free strip $\operatorname{Re} s > 1 - 10^{-6}$ for Hecke L-functions of cyclotomic fields containing the twelfth roots of unity, that every integer other than -1 and the squares is a primitive root modulo infinitely many primes, and that polynomials over $\mathbb{F}_p$ can be factored in deterministic polynomial time; neither bounds $g(p)$.

Remark 1.8. The constant p_0 of Corollary 1.3 is effective but far beyond direct computation. A computation (Section 6) gives $g^*(p) < p$ for every prime $3 \leq p \leq 10^8$, with $\max g^*(p) = 223$ at $p = 73189117$. Erdős's question therefore remains open for the primes in $[10^8, p_0)$.

Remark 1.9. Since $\tau(n) \leq 2^{(1+o(1))\log n/\log\log n}$ [9, Thm. 317], Theorem 1.1 gives $g^*(p) \leq \exp\big(c\log p/\log\log p\big)$ for an absolute constant c. This form is not formalised.

Remark 1.10. Carella has claimed $g^*(p) \ll p^{5/\log\log p}$ [4] and $g^*(p) = O((\log p)^{1+\varepsilon})$ [5, 6] without hypotheses. In [4], the estimate (16) bounds $\sum_{n\le x}\chi(n)\Lambda(n)n^{-2}$ by $O(1/x)$; the sum tends to $-L'(2,\chi)/L(2,\chi)$, which equals $-0.17264\ldots$ for the character modulo 3. In the arXiv version of [5], Lemma 7.1 asserts $E(x,y) \ll yp^{-\varepsilon}$ for the sum $E(x,y)$ of (7.2) when no integer of $[x, x+y]$ is a primitive root. Under that hypothesis every inner sum over m equals -1, so $E(x,y) = -\varphi(p-1)\,\#([x,x+y]\cap\mathbb{Z})/p$ exactly, while $\varphi(p-1)/p \gg 1/\log\log p$; the step (7.2) replaces $\sum_n e(m\tau^n/p)$ by its value at $m=1$. We could not obtain the text of [6].

2. The prime sum of a nonprincipal character

For an arithmetic function f and $x \ge 1$ put

$$S(f;x) = \sum_{n\le x} f(n)\Lambda(n)\Big(1-\frac{n}{x}\Big),$$

and write $S(\chi;x)$ when $f=\chi$. The aim of this section is Proposition 2.6.

Lemma 2.1. *Let $a\colon \mathbb{N}\to\mathbb{C}$ satisfy $\sum_n |a(n)|n^{-2} < \infty$ and put $F(s) = \sum_n a(n)n^{-s}$ for $\operatorname{Re} s \ge 2$. For $x>0$,*

$$\sum_{n\le x} a(n)\Big(1-\frac{n}{x}\Big) = \frac{1}{2\pi}\int_{-\infty}^{\infty} F(2+it)\,\frac{x^{2+it}}{(2+it)(3+it)}\,dt.$$

Proof. For $y>0$ let $k(y) = \frac{1}{2\pi i}\int_{(2)} y^s \frac{ds}{s(s+1)}$. If $y\ge 1$, moving the line to the left through the poles at 0 and -1 gives $k(y) = 1-1/y$; if $y<1$, moving it to the right gives $k(y)=0$. Since $|s(s+1)|^{-1}$ is integrable on $\operatorname{Re} s = 2$ and $\sum_n |a(n)|n^{-2}<\infty$, summation and integration may be exchanged: the right side equals $\sum_n a(n)k(x/n)$. □

Lemma 2.2. *Let χ be nonprincipal modulo q. For $\operatorname{Re} s = \sigma > 0$, $|L(s,\chi)| \le q|s|/\sigma$.*

Proof. Put $A(t) = \sum_{n\le t}\chi(n)$; then $|A(t)|\le q$, and partial summation gives $L(s,\chi) = s\int_1^\infty A(t)t^{-s-1}\,dt$ for $\sigma>0$. Hence $|L(s,\chi)| \le q|s|\int_1^\infty t^{-\sigma-1}dt = q|s|/\sigma$. □

Lemma 2.3. *Let $A_0 = \frac{3}{2}\sum_\ell \ell^{-3/2}$, the sum over primes. For every Dirichlet character χ the series $E(s) = \sum_\ell -\log(1-\chi(\ell)\ell^{-s})$ is holomorphic on $\operatorname{Re} s > 1$, $e^{E(s)} = L(s,\chi)$ there, and $|E(s)| \le A_0$ for $\operatorname{Re} s \ge 3/2$.*

Proof. For $\operatorname{Re} s \ge 1$ we have $|\chi(\ell)\ell^{-s}| \le 1/2$, and $|\log(1-w)| \le \frac{3}{2}|w|$ for $|w|\le 1/2$. The series therefore converges locally uniformly on $\operatorname{Re} s>1$ and is bounded by A_0 on $\operatorname{Re} s \ge 3/2$, and the Euler product gives $e^E = L$. □

Lemma 2.4. *Let $0 < \delta < 1/1000$, $1/2 \le a \le 1$, and let χ be nonprincipal modulo q with $L(s,\chi) \ne 0$ for $\operatorname{Re} s > a + 2\delta$. Put $K_1 = (1 + 2A_0)/(2\delta^2)$. For $a + 8\delta \le \sigma \le 2$ and $t \in \mathbb{R}$,*

$$\Big|\frac{L'}{L}(\sigma + it, \chi)\Big| \le K_1 \log\big(2q(3 + |t|)^2\big).$$

Proof. Let $c = 2+it$, $R = 2-a-3\delta$, $r = 2-a-6\delta$ and $\Lambda_0 = \log(2q(3+|t|)^2) \ge \log 18 > 1$. On the disc $|z-c| < 2-a-2\delta$ we have $\operatorname{Re} z > a+2\delta$, so $L(z,\chi)$ has no zero there and has a holomorphic logarithm h. On $|z-c| < 1/2$ both h and E of Lemma 2.3 are logarithms of L, so they differ by a constant in $2\pi i\mathbb{Z}$; we choose $h = E$ there. Then $|h(c)| \le A_0$.

If $|z - c| \le R$, then $\operatorname{Re} z \ge a + 3\delta \ge 1/2$ and $|z| \le |c| + 3/2 \le 4 + |t|$, so Lemma 2.2 gives $\operatorname{Re} h(z) = \log|L(z,\chi)| \le \log(2q(4 + |t|)) \le \Lambda_0$. The Borel–Carathéodory inequality [18, §5.5] gives, for $|z - c| \le r$,

$$|h(z) - h(c)| \le \frac{2r}{R - r} \max_{|w-c|=R} \operatorname{Re}\big(h(w) - h(c)\big) \le \frac{\Lambda_0 + A_0}{\delta},$$

as $r \le 3/2$ and $R-r = 3\delta$. Hence $|h(z)| \le A_0+(\Lambda_0+A_0)/\delta \le (1+2A_0)\Lambda_0/\delta$ on $|z - c| \le r$. For $w = \sigma + it$ with $a + 8\delta \le \sigma \le 2$ we have $|w - c| = 2 - \sigma \le r - 2\delta$, so the circle $|z - w| = 2\delta$ lies in $|z - c| \le r$, and Cauchy's estimate gives $|L'/L(w,\chi)| = |h'(w)| \le (1 + 2A_0)\Lambda_0/(2\delta^2)$. □

Lemma 2.5. *For $q \ge 1$, $t \in \mathbb{R}$ and $s = \sigma + it$ with $\sigma \ge 1/2$,*

$$\log\big(2q(3 + |t|)^2\big) \le 10 \log(q + 2)(1 + |t|)^{1/2}, \qquad |s(s + 1)| \ge \tfrac{1}{8}(1 + |t|)^2.$$

Proof. We have $\log(2q) \le 2\log(q + 2)$, and $\log y \le 2y^{1/2}$ with $3 + |t| \le 4(1 + |t|)$ gives $\log(3 + |t|) \le 4(1 + |t|)^{1/2}$; the first bound follows from $\log(q + 2) \ge 1$. For the second, $|s| \ge \max(\sigma, |t|) \ge (1 + |t|)/4$ and $|s + 1| \ge \max(\sigma + 1, |t|) \ge (1 + |t|)/2$. □

Proposition 2.6. *Let $0 < \delta < 1/1000$. There is $K = K(\delta)$ such that the following holds. Let $1/2 \le a \le 1$ and let χ be nonprincipal modulo q with $L(s,\chi) \ne 0$ for $\operatorname{Re} s > a + 2\delta$. Then for $x \ge 1$,*

$$|S(\chi; x)| \le K\, x^{a+8\delta} \log(q + 2).$$

Proof. Put $\sigma_0 = a + 8\delta$ and

$$\Phi(s) = -\frac{L'}{L}(s, \chi)\, \frac{x^s}{s(s + 1)}.$$

For $\operatorname{Re} s > 1$, $-L'/L(s,\chi) = \sum_n \chi(n)\Lambda(n)n^{-s}$, so Lemma 2.1 gives $S(\chi; x) = \frac{1}{2\pi}\int \Phi(2+it)\, dt$. The function Φ is holomorphic on $\sigma_0 \le \operatorname{Re} s \le 2$, and by Lemmas 2.4 and 2.5,

$$|\Phi(\sigma + it)| \le 80K_1 \log(q + 2)\, x^{\sigma}(1 + |t|)^{-3/2} \qquad (\sigma_0 \le \sigma \le 2).$$

This tends to 0 as $|t| \to \infty$ uniformly in σ, so Cauchy's theorem on the rectangles $[\sigma_0, 2] \times [-T, T]$, $T \to \infty$, moves the integral to $\operatorname{Re} s = \sigma_0$. Since $\int_{\mathbb{R}} (1+|t|)^{-3/2} dt = 4$, we get $|S(\chi; x)| \le \frac{160 K_1}{\pi} x^{\sigma_0} \log(q+2)$. □

3. Primitive roots as a combination of characters

Fix a prime p, put $N = p-1$, let g generate $(\mathbb{Z}/p\mathbb{Z})^\times$, and let χ_1 be the character modulo p with $\chi_1(g) = e^{2\pi i/N}$. Let χ_0 be the principal character.

Lemma 3.1. *For every integer n,*

$$\mathbf{1}[n \text{ is a primitive root mod } p] = \sum_{d \mid N} \frac{\mu(d)}{d} \sum_{m=0}^{d-1} \chi_1^{mN/d}(n),$$

and $\chi_1^{mN/d} \ne \chi_0$ for $d \mid N$ and $0 < m < d$.

Proof. If $p \mid n$ both sides vanish. Otherwise $n \equiv g^k$, the order of n is $N/(N,k)$, and $\sum_{m<d} e^{2\pi i k m/d} = d\,\mathbf{1}[d \mid k]$; the right side is $\sum_{d \mid (N,k)} \mu(d) = \mathbf{1}[(N,k) = 1]$. The character $\chi_1^{mN/d}$ sends g to $e^{2\pi i m/d} \ne 1$. □

Lemma 3.2. *For $N \ge 1$, $\sum_{d \mid N} \mu(d)/d = \varphi(N)/N \ge 1/\tau(N)$.*

Proof. $\varphi(N)/N = \prod_{r \mid N} (1 - 1/r) \ge 2^{-\omega(N)} \ge 1/\tau(N)$, the product over the prime divisors r of N. □

Lemma 3.3. *For $x \ge 10^4$, $\operatorname{Re} S(\chi_0; x) \ge \frac{1}{2}\psi(x/2) - \log x \ge x/10$.*

Proof. The terms of $\sum_{n \le x} \Lambda(n)(1 - n/x)$ are nonnegative and at least $\Lambda(n)/2$ for $n \le x/2$, so the sum is at least $\psi(x/2)/2$. The terms removed by χ_0 are those with $p \mid n$, that is $n = p^k \le x$, and they contribute at most $\sum_{p^k \le x} \log p \le \log x$. For the second inequality, $\psi(n) \ge n \log 2$ for integers $n \ge 7$ [13], so $\psi(x/2) \ge (x/2 - 1)\log 2$, and $\frac{x}{4}\log 2 - \frac{1}{2}\log 2 - \log x \ge \frac{x}{10}$ for $x \ge 10^4$. □

Lemma 3.4. *If ℓ^k with $k \ge 1$ is a primitive root modulo p, so is ℓ.*

Proof. ℓ is a unit modulo p, and $N = \operatorname{ord}(\ell^k)$ divides $\operatorname{ord}(\ell)$, which divides N. □

4. Proof of Theorem 1.1 and Corollaries 1.2 and 1.3

Proof of Theorem 1.1. Let $\delta = \eta/(2000(8+\eta))$, so $0 < \delta < 1/1000$, and let K be given by Proposition 2.6; put $K' = \max(K, 1)$. Let $\sigma_0 = 7/8 + 8\delta$ and $\beta = 1 - \sigma_0 = 1/8 - 8\delta > 0$. Then

$$(8+\eta)\beta = 1 + \frac{\eta}{8} - \frac{\eta}{250} \ge 1.$$

Put $C_\eta = \max\bigl(10^4, (20K')^{8+\eta}\bigr)$. Given a prime p, let $\tau = \tau(p-1)$, $Y = \tau^2 \log(p+2) \ge 1$ and $x = C_\eta Y^{8+\eta} \ge 10^4$.

By Theorem 1.7, Proposition 2.6 applies with $a = 7/8$ to every nonprincipal χ modulo p, so $|S(\chi; x)| \le B := K' x^{\sigma_0} \log(p+2)$. Let G be the sum $S(f; x)$ with f the indicator of the primitive roots modulo p. By Lemma 3.1,

$$G = c_0 S(\chi_0; x) + \mathcal{E}, \qquad c_0 = \sum_{d|N} \frac{\mu(d)}{d}, \qquad \mathcal{E} = \sum_{d|N} \frac{\mu(d)}{d} \sum_{0<m<d} S(\chi_1^{mN/d}; x),$$

and $|\mathcal{E}| \le \sum_{d|N} \frac{d-1}{d} B \le \tau B$. Lemmas 3.2 and 3.3 give

$$|G| \ge c_0 \operatorname{Re} S(\chi_0; x) - |\mathcal{E}| \ge \frac{x}{10\tau} - \tau B.$$

Now $x^\beta = C_\eta^\beta Y^{(8+\eta)\beta} \ge (20K')^{(8+\eta)\beta} Y \ge 20K'Y$, so $\tau^2 B = K' Y x^{\sigma_0} \le x^{\beta+\sigma_0}/20 = x/20$. Hence $|G| \ge x/(20\tau) > 0$, and some $n \le x$ with $\Lambda(n) \ne 0$ is a primitive root modulo p. Writing $n = \ell^k$, the prime $\ell \le x$ is a primitive root by Lemma 3.4. □

Proof of Corollary 1.2. Take $\eta = 1$ and $\nu = \varepsilon/27$. By [9, Thm. 315] there is D_ν with $\tau(n) \le D_\nu n^\nu$, and $\log(p+2) \le (p+2)^\nu/\nu \le 3^\nu p^\nu/\nu$. Theorem 1.1 gives $\ell \le C_1 (D_\nu^2 3^\nu \nu^{-1})^9 p^{27\nu}$. □

Proof of Corollary 1.3. Corollary 1.2 with $\varepsilon = 1/2$ gives $\ell \le C\sqrt{p} < p$ for $p > C^2$. □

5. The Lindelöf bound and character sums

In this section A_0 is the constant of Lemma 2.3.

Lemma 5.1. *Let $0 < \kappa < 1/1000$ and $\varepsilon > 0$. There is $D = D(\kappa, \varepsilon)$ such that the following holds. Let $1/2 \le a \le 1$ and let χ be nonprincipal modulo q with $L(s, \chi) \ne 0$ for $\operatorname{Re} s > a + 2\kappa$. For $a + 8\kappa \le \sigma \le 2$ and $t \in \mathbb{R}$,*

$$|L(\sigma + it, \chi)| + |L(\sigma + it, \chi)|^{-1} \le D\big(2q(3 + |t|)^2\big)^{\varepsilon}.$$

Proof. Let c, r, Λ_0 and the logarithm h of $L(z, \chi)$ on $|z - c| < 2 - a - 2\kappa$ be as in the proof of Lemma 2.4, with δ replaced by κ. Both h and E are logarithms of L on the disc $|z - c| < 3/4$, where $\operatorname{Re} z > 1$, so $h = E$ there, and $|h(z)| \le A_0$ for $|z - c| \le 1/2$ by Lemma 2.3. The proof of Lemma 2.4 gives $|h(z)| \le M := (1 + 2A_0)\Lambda_0/\kappa$ for $|z - c| \le r$.

Let $w = \sigma + it$, $\rho = |w - c| = 2 - \sigma$, $A_1 = \max(A_0, 1)$ and $\lambda = 4\kappa/(3 \log 3)$; note $M \ge 1$. If $\rho \le 1/2$ then $|h(w)| \le A_0 \le A_1 M^{1-\lambda}$. Otherwise $1/2 < \rho \le r - 2\kappa$, and Hadamard's three circles theorem [18, §5.3] gives $|h(w)| \le A_0^{1-\alpha} M^\alpha$ with $\alpha = \log(2\rho)/\log(2r)$. Since $1 - 6\kappa \le r \le 3/2$,

$$\alpha \le \frac{\log(2r - 4\kappa)}{\log(2r)} = 1 + \frac{\log(1 - 2\kappa/r)}{\log(2r)} \le 1 - \frac{2\kappa/r}{\log 3} \le 1 - \lambda,$$

so again $|h(w)| \le A_1 M^{1-\lambda} = B\Lambda_0^{1-\lambda}$ with $B = A_1((1+2A_0)/\kappa)^{1-\lambda}$. If $\Lambda_0^\lambda \ge B/\varepsilon$ then $B\Lambda_0^{1-\lambda} \le \varepsilon\Lambda_0$, and otherwise $B\Lambda_0^{1-\lambda} \le D_1 := B(B/\varepsilon)^{(1-\lambda)/\lambda}$. Since $\log|L(w,\chi)| = \operatorname{Re} h(w)$,

$$|L(w,\chi)|^{\pm 1} \le e^{|h(w)|} \le e^{D_1} e^{\varepsilon\Lambda_0} = e^{D_1}\big(2q(3+|t|)^2\big)^{\varepsilon},$$

and $D = 2e^{D_1}$ works. □

Lemma 5.2. *Let $|b(n)| \le 1$ for $n \ge 1$ and put $T(y) = \sum_{n\le y} b(n)(y-n)$ for $y > 0$. For $x, h > 0$,*

$$h\Big|\sum_{n\le x} b(n)\Big| \le |T(x+h) - T(x)| + h(h+1).$$

Proof. $T(x+h) - T(x) = h\sum_{n\le x} b(n) + \sum_{x<n\le x+h} b(n)(x+h-n)$, and the last sum has at most $h+1$ terms, each of modulus at most h. □

Proposition 5.3. *Let $0 < \kappa < 1/1000$ and $0 < \varepsilon \le 1/8$. There is $K = K(\kappa,\varepsilon)$ such that the following holds. Let $1/2 \le a$ with $a + 8\kappa \le 1$, and let χ be nonprincipal modulo q with $L(s,\chi) \ne 0$ for $\operatorname{Re} s > a + 2\kappa$. Then for $x \ge 1$,*

$$\Big|\sum_{n\le x} \chi(n)\Big| \le K\, q^{\varepsilon} x^{a+8\kappa+4\varepsilon}.$$

Proof. Put $\sigma_0 = a + 8\kappa \in [1/2, 1]$ and let D be given by Lemma 5.1. Since $2q(3+|t|)^2 \le 18q(1+|t|)^2$, Lemmas 5.1 and 2.5 give, for $s = \sigma + it$ with $\sigma_0 \le \sigma \le 2$,

$$\Big|\frac{L(s,\chi)}{s(s+1)}\Big| \le 8D(18q)^{\varepsilon}(1+|t|)^{-2+2\varepsilon}. \tag{1}$$

Let $T(y) = \sum_{n\le y}\chi(n)(y-n)$. Lemma 2.1 with $a(n) = \chi(n)$ gives $T(y) = \frac{1}{2\pi i}\int_{(2)} L(s,\chi)\, y^{s+1}\frac{ds}{s(s+1)}$. The integrand is entire, and by (1) it tends to 0 as $|t| \to \infty$ uniformly in $\sigma_0 \le \sigma \le 2$, since $2\varepsilon < 1$; Cauchy's theorem moves the line to $\operatorname{Re} s = \sigma_0$. For $h > 0$,

$$T(x+h) - T(x) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \frac{L(s,\chi)}{s(s+1)}\big((x+h)^{s+1} - x^{s+1}\big)\, dt, \qquad s = \sigma_0 + it.$$

Since $(x+h)^{s+1} - x^{s+1} = (s+1)\int_x^{x+h} u^s\, du$ and $|s+1| \le 3(1+|t|)$, its modulus is at most $3(1+|t|)h(x+h)^{\sigma_0}$, and also at most $2(x+h)^{\sigma_0+1}$. The minimum of two numbers is at most the product of their powers $1-4\varepsilon$ and 4ε, so

$$\big|(x+h)^{s+1} - x^{s+1}\big| \le (1+|t|)^{1-4\varepsilon}(x+h)^{\sigma_0}(3h)^{1-4\varepsilon}\big(2(x+h)\big)^{4\varepsilon}.$$

With (1) and $\int_{\mathbb{R}}(1+|t|)^{-1-2\varepsilon}dt = 1/\varepsilon$,

$$|T(x+h) - T(x)| \le \frac{4D(18q)^{\varepsilon}}{\pi\varepsilon}(x+h)^{\sigma_0}(3h)^{1-4\varepsilon}\big(2(x+h)\big)^{4\varepsilon}.$$

Take $h = x^{\sigma_0}$, so $1 \le h \le x$. Then $(x+h)^{\sigma_0} \le 2h$, $(3h)^{1-4\varepsilon} \le 3h$ and $(2(x+h))^{4\varepsilon} \le 4x^{4\varepsilon}$, so the right side is at most $\frac{1728D}{\pi\varepsilon} q^{\varepsilon} h^2 x^{4\varepsilon}$. By Lemma 5.2 with $b = \chi$,

$$\Big|\sum_{n\le x} \chi(n)\Big| \le \frac{1728D}{\pi\varepsilon} q^{\varepsilon} h x^{4\varepsilon} + h + 1 \le \Big(\frac{1728D}{\pi\varepsilon} + 2\Big) q^{\varepsilon} x^{\sigma_0+4\varepsilon}. \qquad \square$$

Proof of Theorem 1.4. Let $\kappa = \min(\delta/8, 1/2000)$ and $a = 7/8 - 2\kappa$, so $1/2 \le a \le 1$ and, by Theorem 1.7, $L(s,\chi) \ne 0$ for $\operatorname{Re} s > a + 2\kappa$. Let D be given by Lemma 5.1 with $\varepsilon = \delta/2$. If $7/8+\delta \le \sigma \le 2$, then $\sigma \ge a + 8\kappa$, and $2q(3+|t|)^2 \le (2q(3+|t|))^2$ gives $|L| + |L|^{-1} \le D 2^{\delta} (q(3+|t|))^{\delta}$. If $\sigma > 2$, Lemma 2.3 gives $|L|^{\pm 1} \le e^{A_0}$. So $C_\delta = D2^{\delta} + 2e^{A_0}$ works. $\square$

Proof of Theorem 1.5. Let $\delta' = \min(\delta, 1)$, $\kappa = \min(\delta'/12, 1/2000)$, $a = 7/8 - 2\kappa$ and $\varepsilon = \delta'/8$. Then $a + 8\kappa = 7/8 + 6\kappa \le 1$, and Proposition 5.3, with Theorem 1.7, gives the bound $K q^{\delta'/8} x^{7/8+6\kappa+\delta'/2} \le K q^{\delta} x^{7/8+\delta}$. $\square$

Proof of Corollary 1.6. Let $\delta = (\theta - 7/8)\eta/(1+\eta)$, so that $\delta/\eta + 7/8 + \delta = \theta$. For $x \ge q^{\eta}$ we have $q^{\delta} \le x^{\delta/\eta}$, and Theorem 1.5 gives $|\sum_{n\le x} \chi(n)| \le C_\delta x^{\theta}$. With $\theta = 15/16$ this is $o(x)$. $\square$

6. Formal verification and computations

The Lean 4 development (Lean v4.34.1, Mathlib v4.34.1) proves

```
theorem least_prime_primitive_root_bound {η : ℝ} (hη : 0 < η) :
    ∃ C : ℝ, 0 < C ∧ ∀ p : ℕ, p.Prime → ∃ ℓ : ℕ, ℓ.Prime ∧
      orderOf (ℓ : ZMod p) = p - 1 ∧
      (ℓ : ℝ) ≤ C * (((p - 1).divisors.card : ℝ) ^ 2 *
        Real.log (p + 2)) ^ (8 + η)
theorem least_prime_primitive_root_le_rpow {ε : ℝ} (hε : 0 < ε) :
    ∃ C : ℝ, 0 < C ∧ ∀ p : ℕ, p.Prime → ∃ ℓ : ℕ, ℓ.Prime ∧
      orderOf (ℓ : ZMod p) = p - 1 ∧ (ℓ : ℝ) ≤ C * (p : ℝ) ^ ε
theorem erdos_985_of_large : ∃ p₀ : ℕ, ∀ p : ℕ, p.Prime →
    p₀ ≤ p → ∃ q : ℕ, q.Prime ∧ q < p ∧
      orderOf (q : ZMod p) = p - 1
```

which are Theorem 1.1 and Corollaries 1.2 and 1.3, and

```
theorem lindelof {w : ℝ} (hw : 0 < w) :
    ∃ C : ℝ, 0 < C ∧ ∀ (q : ℕ) [NeZero q] (χ : DirichletCharacter ℂ q),
      χ ≠ 1 → ∀ s : ℂ, 7 / 8 + w ≤ s.re →
        ‖DirichletCharacter.LFunction χ s‖ +
          ‖(DirichletCharacter.LFunction χ s)⁻¹‖ ≤
          C * ((q : ℝ) * (3 + |s.im|)) ^ w
theorem character_sum_bound {w : ℝ} (hw : 0 < w) :
    ∃ C : ℝ, 0 < C ∧ ∀ (q : ℕ) [NeZero q] (χ : DirichletCharacter ℂ q),
      χ ≠ 1 → ∀ x : ℝ, 1 ≤ x →
        ‖∑ n ∈ Finset.Icc 1 ⌊x⌋₊, χ n‖ ≤
          C * (q : ℝ) ^ w * x ^ (7 / 8 + w)
theorem character_sum_le_rpow {η θ : ℝ} (hη : 0 < η) (hθ : 7 / 8 < θ) :
```

```
∃ C : ℝ, 0 < C ∧ ∀ (q : ℕ) [NeZero q] (χ : DirichletCharacter ℂ q),
  χ ≠ 1 → ∀ x : ℝ, 1 ≤ x → (q : ℝ) ^ η ≤ x →
    ‖∑ n ∈ Finset.Icc 1 ⌊x⌋₊, χ n‖ ≤ C * x ^ θ
```

(namespace `Vinogradov.CharSum`), which are Theorems 1.4 and 1.5 and Corollary 1.6 (there $x \geq 1$ follows from $x \geq q^{\eta}$). Theorem 1.7 enters as the theorem

`OAI.DirichletCharacter.LFunction_ne_zero_of_seven_eighths_lt_re`

of [14], stated for Mathlib's `DirichletCharacter.LFunction` and compiled from its source. The Lean proofs of Lemmas 2.4 and 5.1 use the logarithmic control lemma of [14] in place of the Borel–Carathéodory and three circles arguments, and the proof of Lemma 3.3 uses Mathlib's bound $\psi(y) \geq (y - 1)\log 2 - \log(y + 2)$ in place of [13]. Every theorem depends only on the axioms `propext`, `Classical.choice` and `Quot.sound`.

Programs in Python, C, C++ and Julia compute $g^*(p)$ for all primes $3 \leq p \leq 10^8$ (Remark 1.8), check Lemma 3.1 numerically for $p \leq 200$, the inequality of Lemma 3.3 on $10^4 \leq x \leq 2 \cdot 10^7$, Lemma 3.2 for $N \leq 10^7$, the exponent identity in the proof of Theorem 1.1 in exact arithmetic, the two values in Remark 1.10, the identity in the proof of Lemma 5.2 in exact arithmetic for six characters, the inequalities in the proof of Proposition 5.3 at $2 \cdot 10^5$ random points, and the choices of parameters in Section 5 in exact arithmetic.

Data availability

The Lean project, the programs and a script that runs them are archived at Zenodo, doi:10.5281/zenodo.0000000.

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